Infinitesimal Calculus 2, list of theorem for Quiz 1

1. Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Let $(n_k)_{k=1}^{\infty}$ be a strictly increasing sequence of natural numbers. Suppose that the sequence $(a_n)_{n=1}^{\infty}$ converges in the extended sense. Prove that the sequence $\left(a_{n_k}\right)_{k=1}^{\infty}$ converges in the exended sense and
$$\lim_{k\to\infty} a_{n_k} = \lim_{n\to\infty} a_n$$

2. Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers. Suppose $(a_{2k})_{k=1}^{\infty}$ and $(a_{2k-1})_{k=1}^{\infty}$ both converge in the extended sense, and $\lim\limits_{k\to\infty} a_{2k} = \lim\limits_{k\to\infty} a_{2k-1}$. Prove that the sequence $(a_n)_{n=1}^{\infty}$ converges in the extended sense, and
$$\lim_{n\to\infty} a_n = \lim_{k\to\infty} a_{2k} = \lim_{k\to\infty} a_{2k-1}.$$

3. Let f, F, G be three functions defined on an interval I. Suppose that F, G are anti-derivatives of f. Prove $\exists C \in \mathbb{R}$ such that
$$F(x) = G(x) + C, \qquad \forall x \in I.$$

4. Let $a, b \in \mathbb{R}$ such that $a < b$ and let $C \in \mathbb{R}$. Prove that the function $f(x) = C$ for every $x \in [a, b]$ is integrable on $[a, b]$, and
$$\int_a^b C\,dx = C \cdot (b - a).$$

5. Prove that the Dirichlet function is not integrable on any closed interval $[a, b] \subseteq \mathbb{R}$.